Homework 3

Assigned: Session 5, October 24, 2026

Due: Session 7, November 7, 2026

Covers: Lectures 5 and 6

Instructions: Work by hand; no programming is required. State the conventions you use (qubit order, basis order, gate order) and show intermediate states rather than only final answers. Where a question asks for an explanation, two or three precise sentences are expected. Part A can be attempted after Lecture 5 and Part B after Lecture 6. Estimated total time: two to three hours. Submission arrangements are announced separately.

Conventions: as in Homework 1. In addition, O_f denotes a phase oracle, $|s\rangle$ the uniform superposition, and $D = 2\,|s\rangle\langle s| - \mathbb{I}$ the Grover diffusion operator.

A Quantum search

Attempt after Lecture 5.

Problem A1. Grover search on eight items

Let $N = 8$ with a single marked item w. Let $|s\rangle$ be the uniform superposition, O_f the phase oracle that negates $|w\rangle$, and $D = 2\,|s\rangle\langle s| - \mathbb{I}$.

(a) Give $\sin\theta$ and θ in degrees, where θ is the angle between $|s\rangle$ and the unmarked subspace.

(b) Using $P_k = \sin^2((2k+1)\theta)$, compute the success probability after $k = 0, 1, 2, 3$ iterations. Which k is optimal, and how does it compare with $\frac{\pi}{4}\sqrt{N}$?

(c) Verify P_1 directly by inversion about the mean: write the eight amplitudes after O_f, compute their mean, reflect each amplitude through the mean, and read off the marked probability.

(d) What happens if the algorithm is run for $k = 3$ iterations? Explain in one sentence.

Problem A2. Two marked items

Now let $N = 8$ with $M = 2$ marked items, and let $|w\rangle$ denote the normalized uniform superposition of the two marked states.

(a) Give $\sin\theta$ and θ, and compute the probability of measuring a marked item after one iteration.

(b) Compute the same probability after two iterations. Compare with the single-marked case and comment.

(c) An experimenter believes there is one marked item when in fact there are two, and therefore runs the number of iterations that is optimal for $M = 1$. What success probability do they obtain? What does this say about the need to know M?

Problem A3. What the quadratic speedup means

A symmetric cipher has a 40-bit key (a toy size chosen for arithmetic). Given one plaintext-ciphertext pair, define $f(k) = 1$ if key k encrypts the plaintext to the given ciphertext and 0 otherwise, and assume exactly one key does.

(a) How many evaluations of the cipher does a classical brute-force search need on average? How many Grover iterations, each containing one evaluation of the cipher as a reversible circuit, does a quantum search need?

(b) Express the quantum count as an effective classical key length: the quantum search is as hard as a classical search over how many bits?

(c) If the cipher evaluation costs c elementary operations classically and c_q elementary quantum gates as a reversible circuit, write the two total costs. Why does the comparison depend on c_q/c, and why is c_q typically much larger than c?

(d) Explain in two sentences why the same reasoning does not give a quantum advantage for finding a name in an alphabetically sorted list.

B Noise and error correction

Attempt after Lecture 6.

Problem B1. Coherences, purity, and dephasing

Let $|\psi\rangle = \frac{\sqrt{3}}{2}|0\rangle + \frac{1}{2}|1\rangle$, let $\rho_1 = |\psi\rangle\langle\psi|$, and let $\rho_2 = \frac{3}{4}|0\rangle\langle 0| + \frac{1}{4}|1\rangle\langle 1|$.

(a) Write ρ_1 and ρ_2 as 2×2 matrices. Which entries agree and which differ?

(b) Compute $\mathrm{tr}(\rho P_+)$ for both, where $P_+ = |+\rangle\langle +|$, using $\mathrm{tr}(\rho P_+) = \frac{1}{2} + \mathrm{Re}\,\rho_{01}$. Which measurement distinguishes the two preparations?

(c) Compute $\mathrm{tr}(\rho^2)$ for both. What does this quantity detect that the coherences alone do not?

(d) Apply the dephasing channel $\rho \mapsto (1-p)\rho + p\,Z\rho Z$ to ρ_1 and write the result. For which p does ρ_1 become ρ_2?

Problem B2. The three-qubit code in action

A logical qubit $\alpha|0\rangle + \beta|1\rangle$ is encoded as $|\psi_L\rangle = \alpha|000\rangle + \beta|111\rangle$. The syndrome is the pair of outcomes of measuring Z_1Z_2 and Z_2Z_3.

(a) A bit flip X_2 occurs. Write the state, give the syndrome, and give the recovery operation. Verify that recovery restores $|\psi_L\rangle$.

(b) Two bit flips X_1X_3 occur. Write the state and give the syndrome. What recovery does the decoder apply, and what is the resulting state? Express the result as a logical operation on $\alpha|0\rangle + \beta|1\rangle$.

(c) A phase flip Z_2 occurs. Write the state and give the syndrome. What does the decoder do, and what has happened to the logical qubit?

(d) Explain in two sentences why the syndrome measurement in (a) reveals nothing about α and β.

Problem B3. When the code helps

In the restricted model, each physical qubit independently suffers a bit flip with probability p, and the three-qubit code fails when two or more qubits flip.

(a) Show that the failure probability is $p_{\text{fail}} = 3p^2 - 2p^3$.

(b) Evaluate p_{fail} for $p = 0.05$ and for $p = 0.6$. In which case does encoding help, and by what factor?

(c) Show that, for $0 < p < 1$, $p_{\text{fail}} < p$ exactly when $p < \frac{1}{2}$, and interpret this threshold in one sentence.

(d) The analysis assumes that encoding, syndrome measurement, and recovery are themselves perfect. Name one way in which a fault in the syndrome measurement could cause the procedure to introduce an error rather than remove one.